Team D: Evaluation (Justyna, Mario)

Task 1 (PARALLEL after Definition Pack): Create the scoring sheet
One shared scoring format (simple yes/no):

- *Evidence relevance: are Top 5 items relevant*
- *Mapping accuracy: label matches reference*
- *Citation validity: citation supports the claim*

Deliverable: one scoring sheet template + scoring instructions.

Task 2 (SEQUENTIAL after scoring sheet + mapping rubric exist): Create the 15-vignette pack
15 NSCLC vignettes with:

- *inputs filled*
- *the clinical question type*
- *the expected "gold" mapping labels (based on the guideline excerpt set)*
  Validation plan:
- *Federico reviews relevance and mapping for at least a subset*
- *team reviews citation validity*

Deliverable: "15 Vignettes Pack v1."

TEAM D CONTRIBUTION BELOW ⬇️

I. Two-Layer Architecture: System Integrity & Clinical Correctness

Purpose

The evaluation framework separates **system functionality** from **clinical correctness** to ensure clarity, transparency, and defensible error analysis.

This layered design ensures that:

- Operational failures are distinguished from clinical misclassifications.
- Deterministic system outputs are not confused with evaluation judgments.
- Errors are measurable, interpretable, and attributable to specific stages of the pipeline

Two layer Architecture"

- Layer 1 — Did the system function?
- Layer 2 — Was it correct?

Inside Layer 2:

1. Clinical relevance correctness
2. Guideline mapping correctness
3. Citation correctness
4. Deterministic model logic correctness

Layer 1 - Technical Layer (System Functionality Checks)

This layer verifies whether the system operates structurally as intended.
These checks assess system integrity, not medical accuracy.

Key Principle
A “Yes” in this layer means the system functioned.
It does not imply clinical correctness.

Step 1: Input Acceptance

Question: Does the system successfully accept structured vignette data?

- Yes = Required fields parsed and stored
- No = Input rejected or improperly processed

Failure here indicates a **system-level failure**, not a medical error.

Step 2: Evidence Retrieval

Question: Does the system produce medical evidence output?

- Yes = Evidence items are generated
- No = No evidence retrieved

This check ensures system functionality only.
It does **not** assess relevance, ranking or correctness.

Step 3: Label Assignment

Allowed labels

- Aligned (supports direction already reflected in guideline)
- Guideline-silent (not covered or not yet updated),
- Conflict (suggests direction that contradicts guideline)

Question: Does the system assign exactly one valid label (out of the three required labels)?

- Yes = One valid label assigned
- No = Missing, multiple, or invalid label

This ensures structural completeness of output.

Layer 2 - Evaluation Layer (Clinical & Logical Correctness Checks)

This layer evaluates whether the system's outputs are **correct relative to predefined reference standards** developed by Teams B and C. Evaluation layer 2 may be started only after all stages in Technical Layer 1 are resolved as "Yes" to avoid data pollution and evaluation noise.

Evaluation judgments use:

- Correct
- Error

Step 1: Relevance Assessment

Relevant if evidence applies to:

- NSCLC (project scope)
- Disease stage
- ECOG (MVP slice)
- Biomarkers
- Treatment question

Question: Is the produced evidence relevant to the patient vignette?

- Correct = System classification matches reference relevance
- Error = Misclassification (False Positive or False Negative)

Step 2: Guideline Mapping Accuracy

Question: Is the assigned label (Aligned / Silent / Conflict) correct relative to the reference guideline interpretation?

- Correct = Label matches predefined reference mapping
- Error = Incorrect mapping or incorrect Silence

Human-in-the-loop validation is required at this stage for Phase 2.

Step 3: Citation Validity

Question: Does the citation provided support the claim made?

- Correct = Citation directly supports output
- Error = Citation missing, irrelevant, or unsupported

Step 4: Model Logic Validation (Deterministic Rule Audit)

This step evaluates whether the deterministic model rules (Team C specification) were executed correctly.
This is not clinical correctness - it is logical fidelity to the model specification.

This step audits:

4A. Clinical Relevance Gate

Question:

- Was gate logic applied correctly as specified?
- Were biomarker, histology, and disease filters enforced exactly as defined?

Checks:

- Disease match enforcement
- Setting compatibility rules
- Histology compatibility rules
- Biomarker gating logic
- Correct handling of “mixed” and “unspecified” cases

Evaluation:

Correct = Gate behavior matches model specification
Error = Incorrect inclusion or exclusion

4B. Evidence Ranking Score (ERS) Calculation Audit

ERS = Evidence Ranking Score (0–60)

Question: Was ERS computed exactly according to the predefined A/B/C/D scoring rules?

Audit checks:

- Correct assignment of Evidence Strength points (A)
- Correct Dataset Robustness scoring (B)
- Correct Source Credibility scoring (C)
- Correct Recency scoring (D)
- Correct handling of null/default values
- Correct total score computation
- Correct application of threshold (≥30 for “Top Evidence”)
- Correct deterministic tie-break logic

Evaluation:

- Correct = Score and ranking match deterministic specification
- Error = Miscalculated component, incorrect total, threshold misapplication, or tie-break violation

4C. Ranking Consistency & Invariance Validation

Purpose:
Verify that the ERS ranking behaves consistently with defined monotonic scoring rules.
Define explicit invariants, for example:

1. If Evidence A has higher A-score (Evidence Strength) than Evidence B and all other components equal, ERS_A ≥ ERS_B.
2. Increasing sample size (N) must not decrease ERS.
3. More recent publication year must not decrease ERS.
4. Higher total ERS must always rank above lower ERS (no sorting inconsistencies).
5. Threshold rule (≥30) must always classify evidence correctly into Top vs Secondary.

Evaluation:
- Correct = All ranking invariants satisfied.
- Error = Any invariant violated + description of error.

4D. Label Logic Consistency

- Was the label derived strictly according to guideline stance logic?
- Were “aligned,” “conflict,” and “silent” assigned per specification?

Correct = Label matches deterministic mapping logic
Error = Logical misapplication of stance rules

Error Analysis Framework

Errors are calculated **only in the Evaluation Layer**, not in the Technical Layer that should be fully operable before Evaluation Layer starts.

The Evaluation Layer compares system outputs against a **predefined reference standard**, which is frozen prior to testing and varies by evaluation step (vignette definition, expert-reviewed relevance annotations, guideline mapping table, citation verification).

Step	What Is Being Judged	Required Reference
Vignette interpretation	Input mapping correctness	Frozen vignette definition
Evidence relevance	Applicability to patient	Expert-reviewed relevance annotation
Guideline alignment	Label correctness	Predefined guideline mapping table
Citation validity	Traceability	Human verification of citation support

1. Evidence Relevance Classification

<u>Reference Standard</u>

Relevance annotations are defined using:
- The frozen vignette definition

- Expert-reviewed relevance mapping (Team B + clinical validation)

<u>For relevance classification:</u>

- False Positive (FP): System marks Relevant but reference says Not Relevant
- False Negative (FN): System marks Not Relevant or Silent but reference says Relevant
- Abstention (Silent Case): System marks *Silent* but reference says *Relevant*. Abstentions are tracked separately from FN.
  They are not automatically counted as False Negatives but may be included in a combined "Miss Rate" if reported.

2. Guideline Alignment Classification

<u>Reference Standard</u>

Guideline alignment is determined using a predefined mapping table based on the selected guideline database (ESMO), validated prior to evaluation.

<u>For guideline alignment classification:</u>

- False Positive (FP): System marks Aligned but reference says Conflict or Guideline-Silent
- False Positive (FN): System marks Conflict or Guideline-Silent but reference says Aligned

3. Domain risk asymmetry:

In clinical decision-support systems False Negatives (missed relevant evidence) carry higher clinical risk than False Positives. Therefore, minimizing FN is prioritized over maximizing precision. The primary optimization objective is to Maximize Recall for the Relevant class.

- Recall (Relevant class) is prioritized
- Target FN rate ≤ 5% (MVP validation goal)
- False Positive Rate ≤ 10–15%

<u>II. Performance Metrics (Classification-Based)</u>

Performance evaluation is divided into two domains:

1 . Clinical Classification Performance
2. Deterministic Model Integrity Performance

These domains must not be combined.

1. Clinical Classification Metrics

(Reference-Based Evaluation — Layer 2 Steps 1–3)

Because the system produces categorical outputs
(Relevant / Not Relevant / Silent, Aligned / Silent / Conflict),
evaluation uses classification metrics, not regression metrics.

A. Core Classification Error (0–1 Loss)

Misclassification Rate
"How often did the system get the label wrong?
Misclassification Rate = Number of incorrect predictions/ total prediction
or
Misclassification Rate = (FP+FN)/Total
This represents overall classification error and aligns directly with generalization theory.
Primary generalization metric.

Classification 0-1 loss (defines what error is)
0 = if prediction is correct
1= if prediction is incorrect

We use 0–1 loss as our primary classification risk metric because it directly measures misclassification and aligns with generalization theory. The chosen performance target reflects MVP constraints, not an expectation of near-zero error

B. Confusion Matrix (Domain-Specific Matrices)

Because the system performs multiple distinct classification tasks, confusion matrices are maintained separately for each evaluation domain to preserve interpretability and diagnostic value.

B1. Evidence Relevance (Binary Classification)
A standard 2×2 confusion matrix is used:

	Reference Relevant	Reference Not Relevant
Predicted Relevant	True Positive (TP)	False Positive (FP)
Predicted Not Relevant	False Negative (FN)	True Negative (TN)

This matrix is used to compute:

- Misclassification rate (0–1 loss)
- False Positive Rate (FP ≤ 10–15%)
- False Negative Rate (FN ≤ 5%)
- Recall (Recall ≥ 95%)
- Precision
- F1 Score

This matrix directly reflects clinical risk asymmetry, particularly the impact of False Negatives.

B2. Guideline Mapping (Multi-Class Classification)

Because mapping involves three labels: Aligned, Guideline-Silent, Conflict - so a 3×3 confusion matrix is used:

	Reference Aligned	Reference Silent	Reference Conflict
Predicted by the system - Aligned	Correct	Error	Error
Predicted by the system - Silent	Error	Correct	Error
Predicted by the system - Conflict	Error	Error	Correct

This preserves semantic distinctions between Silent and Conflict, which represent different clinical meanings. Collapsing these categories into binary labels would mask important diagnostic information and reduce interpretability.

✔ Overall mapping error ≤ 10–15%
✔ Report per-class error rates
✔ Optional: Aligned-specific FN/FP if clinically justified

B3. Citation Validity (Binary Classification: Valid/Invalid)

Citation validity evaluates whether the system's cited sources directly support the specific claim made in the output. This assessment is performed independently of relevance and guideline mapping.

Reference Standard

For each output claim, a predefined reference expectation must specify:

- Whether a citation is required (Yes / No)
- The type of citation expected (Evidence study / Guideline source)
- Whether the cited source text supports the specific claim

Citation validity is evaluated against this frozen reference.

Evaluation is performed only for cases where a citation is expected.

	Reference Valid	**Reference Invalid**
Predicted Valid	True Positive (TP)	False Positive (FP)
Predicted Invalid	False Negative (FN)	True Negative (TN)

Where:

- True Positive (TP):
  Citation is present and directly supports the claim.
- False Positive (FP):
  Citation is present but does not support the claim (misleading or irrelevant).
- False Negative (FN):
  Citation is required but missing.
- True Negative (TN):
  Citation correctly omitted when not required.

Additional Operational Definitions

A citation is considered **Valid** only if:

1. The cited document matches the claim's intervention and population context.
2. The cited text supports the direction of the claim (benefit / recommendation / stance).
3. The source is correctly attributed (no mismatch between citation ID and referenced content).

A citation is considered **Invalid** if:

- It is unrelated to the claim.
- It does not support the claim direction.
- It cites a general topic but not the specific intervention/population.
- It is missing when required.

Reported Metrics

- Citation Error Rate
- Missing Citation Rate (FN / Required Cases)
- Misleading Citation Rate (FP / Provided Citations)

Target (MVP): ≤ 10–15% overall citation error
Deterministic citation logic (if rule-based): 100% fidelity expected.

Important Principle

Citation validity evaluates **traceability and justification**, not clinical truth.
A claim may be clinically correct but still receive a citation error if the supporting source is missing or improperly matched.

Principle

Confusion matrices are not aggregated across domains.
Each classification task is evaluated independently to prevent dilution of error patterns and to enable targeted system improvement.

C. False Positive Rate (FP)

The system marks evidence as relevant/aligned when it is not.
FPR = FP/{FP + TN}
Target (MVP): ≤ 10–15%

D. False Negative Rate (FN)

The system fails to mark relevant evidence as relevant.
FNR = FN/{TP + FN}
Target (MVP): ≤ 5%

In clinical decision support, FN is prioritized due to higher risk of missed relevant evidence.

E. Recall (Sensitivity – Relevant Class)

Recall = TP/{TP + FN}
Measures ability to capture all truly relevant evidence.

Target: ≥ 95%
Primary optimization metric.

F. Optional diagnostics:

- Precision: Measures reliability of “Relevant” predictions.
  Precision = {TP}/{TP + FP}
- F1 Score: Balances precision and recall.Recommended reporting metric: **Recall + F1**
  F1 = 2 *{Precision *Recall}/{Precision + Recall}
- Overall Accuracy**.** Note: Accuracy alone is insufficient in imbalanced datasets.
  Accuracy = {TP + TN}/{Total}

2. Deterministic Model Integrity Metrics

(Layer 2 – Step 4)
These metrics evaluate implementation fidelity relative to the frozen model specification.
Target for all deterministic logic metrics: 100% fidelity.

A. Clinical Relevance Gate Accuracy

% of cases where gate inclusion/exclusion matches specification.

B. ERS (Evidence Ranking Score) Calculation Accuracy

% of cases where:

- A/B/C/D components are correctly assigned
- Total ERS computed correctly
- Threshold (≥30) applied correctly
- Tie-break logic applied correctly

C. Ranking Invariance Satisfaction Rate

% of ranking invariants satisfied (e.g., higher A score must not rank below lower A when other factors equal).

D. Topic Mapping Accuracy

% of cases where topic selection matches deterministic overlap rules.

E. Label Logic Accuracy

% of cases where aligned / conflict / silent labels match stance rules.

3. Correctness Validation Strategy

Generalization Control — Applies to Clinical Metrics Only)
To ensure generalization and avoid overfitting

A. Data Split

- Small dataset (≤30 vignettes):
  - Leave-One-Out Cross-Validation (LOOCV)
    OR
  - 70% Train / 30% Test split

Rules:

- No overlap between training and testing vignettes
- Threshold tuning only on training/validation data
- Test set used once for final evaluation

B. Probability Threshold (if applicable)

Only relevant if the model outputs probabilities.

Default:

- Threshold = 0.5

If prioritizing low FN:

- Slightly lower threshold (e.g., 0.4)
- Must be tuned using validation data only

No arbitrary thresholds without ROC-based justification.

C. Regularization (Polynomial Degree 6)

Given:

- 6 input features
- Polynomial expansion (degree 6)
- Small dataset

Risk: High variance / overfitting

Recommended:

- Ridge (L2) regularization

Why:

- Penalizes large coefficients
- Reduces variance
- Improves generalization error
- Stabilizes high-degree polynomial models

Regularization strength should be tuned using cross-validation.

Note: Why NOT Use MSE, MAE, or R^2

MSE, MAE, and R^2 are regression metrics, used when predicting continuous values.
Your system outputs:

- Categorical labels
- Discrete decisions

There is no meaningful numeric distance between:

- Aligned vs Conflict
- Relevant vs Not Relevant

Regression metrics assume:

- Continuous target
- Numeric distance is meaningful

Classification evaluation instead requires:

- Confusion matrix
- Recall
- Precision
- F1 score
- FP/FN analysis

Therefore:
MSE, MAE, and R^2 are inappropriate because the task is classification, not regression.

III. Governance Statement

This evaluation framework:

- Separates system integrity, model logic fidelity, and clinical correctness
- Uses classification error aligned with generalization theory
- Explicitly measures misclassification (0–1 loss)
- Adds deterministic model integrity validation
- Incorporates bias–variance controls (data split + regularization)
- Avoids inappropriate regression metrics
- Prioritizes minimizing False Negatives due to medical risk asymmetry

Performance Metrics Overview (Multi-Matrix Architecture)

Domain	Classification Structure	Metrics Reported	MVP Target	Purpose
Evidence Relevance	**Binary (2×2 Confusion Matrix)** Relevant / Not Relevant	Misclassification Rate (0–1 loss) False Negative Rate (FN) False Positive Rate (FP) Recall Precision F1 Score Accuracy	FN ≤ 5% FP ≤ 10–15% Overall ≤ 10–15%	Control clinical omission risk (missed relevant evidence prioritized)
Guideline Mapping	**Multi-Class (3×3 Confusion Matrix)** Aligned / Silent / Conflict	Overall Mapping Error Rate Per-Class Error Rates (Aligned, Silent, Conflict) Optional Binary View (Aligned vs Not Aligned)	Overall ≤ 10–15%	Preserve semantic distinctions between alignment, silence, and conflict without collapsing meaning
Citation Validity	**Binary (2×2 Confusion Matrix)** Valid / Invalid	Citation Error Rate Missing Citation Rate (FN) Misleading Citation Rate (FP)	≤ 10–15%	Ensure traceability and justification independent of clinical mapping
Model Logic – Gate	Rule Audit (Binary)	Gate Accuracy Rate	100%	Validate deterministic eligibility logic
Model Logic – ERS Calculation	Rule Audit (Binary)	ERS Component AccuracyTotal Score AccuracyThreshold Integrity	100%	Ensure Evidence Ranking Score fidelity
Model Logic – Ranking Invariance	Rule Audit (Binary)	Invariance Satisfaction Rate	100%	Ensure monotonic and stable ranking behavior
Model Logic – Topic & Label Logic	Rule Audit (Binary)	Mapping Logic AccuracyLabel Logic Accuracy	100%	Validate deterministic stance assignment

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